

CASE 36

SYSTEMIC LUPUS ERYTHEMATOSUS

A disease caused by immune complexes.

Immune complexes are produced whenever there is an antibody response to a soluble antigen. As the immune response progresses, larger immune complexes form that trigger the activation of complement. Activated complement components then bind the triggering immune complexes. Large complexes are efficiently cleared by binding to complement receptor 1 (CR1) on erythrocytes, which transfer the immune complexes to the liver and spleen. There, they are removed from the red-cell surface through interaction with a variety of complement and Fc receptors on Kupffer cells and other phagocytes (Fig. 36.1). When antigen is released repeatedly, there may be a sustained formation of small immune complexes that tend to become trapped in the small blood vessels of the renal glomeruli and synovial tissue of the joints.

The most common immune-complex diseases are listed in Fig. 36.2. In subacute bacterial endocarditis, bacteria reside for a protracted period on the heart valves. This infection and subsequent inflammation damage the valve. At the same time, the antibody response to the prolonged presence of the bacteria is intense, and immune complexes of IgG antibodies and bacterial antigens are formed. These complexes become trapped in the renal glomeruli and cause glomerulonephritis. The immunoglobulins in the immune complexes provoke the formation of anti-IgG IgM antibodies known as rheumatoid factor (see Case 35). In a similar fashion, viral hepatitis can become a chronic infection that provokes a marked IgG antibody response, with the consequent formation of virus-containing immune complexes and rheumatoid factor. The immune complexes can become entrapped in the renal glomeruli as well as in small blood vessels of the skin, nerves, and other tissues, where they cause inflammation of the blood vessels (vasculitis). The antibodies in the virus-containing immune complexes precipitate in the cold (less than 37°C) and are therefore termed cryoglobulins.

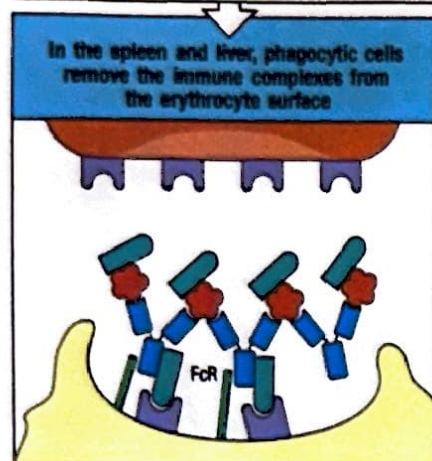
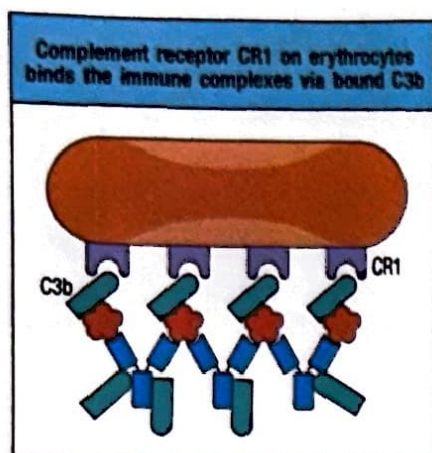
The typical immune-complex disease is systemic lupus erythematosus (SLE), which is characterized by the formation of antibodies against DNA and other nuclear antigens. Every day, millions of nuclei are extruded from erythroblasts in the bone marrow as they mature into red blood cells (erythrocytes). This event, among others, provides a rich source of DNA in those individuals prone to mounting an immune response to DNA and subsequently developing SLE.

TOPICS BEARING ON THIS CASE:

Clearance of immune complexes by complement

Immune-complex disease

Coombs tests



Sixteen-year-old girl, butterfly rash and symmetric morning stiffness.

Joint pain worse, enlarged lymph nodes. Prescribe prednisone and naproxen.

Fig. 36.2 Three autoimmune diseases that result in damage by immune complexes.

Fig. 36.1 Immune complexes are cleared from the circulation by binding to complement and Fc receptors. Immune complexes activate C3 in the serum, and bind activated complement components C3b, C4b, and C2a. C3b binds to complement receptors on erythrocytes, which transport the immune complexes to the spleen and liver, where complement receptors and Fc receptors on phagocytic cells bind to complement components and to the Fc portion of antibodies, and are thereby stimulated to engulf the complexes and degrade them.

The case of Nicole Chawner: too much sun at the beach.

Nicole Chawner was a healthy 16-year-old until this summer. A few days after excessive exposure to the sun on the beach, Nicole developed a red rash on her cheeks. She saw her family doctor, who recognized that the butterfly rash on her cheeks and bridge of her nose was typical of systemic lupus erythematosus (SLE) (Fig. 36.3).

He referred Nicole to the Children's Hospital, where she was asked about any other problems she might have noticed. Nicole said that when she woke up in the morning her fingers and knees were stiff, although they became better as the day progressed. Nicole had also noticed some symmetric swelling in her fingers.

A blood sample was taken from Nicole to ascertain whether she had anti-nuclear antibodies (ANA). These were positive, at a titer of 1:1280 (low titers of ANA are found in more than 10% of otherwise healthy people). Because of this result, further tests were performed for antibodies characteristically found in SLE. An elevated titer of antibodies against double-stranded DNA was also found. Her serum C3 level was 73 mg dl^{-1} (normal $100\text{--}200 \text{ mg dl}^{-1}$). Her platelet count was normal at $225,000 \mu\text{l}^{-1}$, and her direct and indirect Coombs tests were negative, as was a test for anti-phospholipid antibodies. A urine sample was also found to be normal.

Nicole was advised to take an antimalarial agent, hydroxychloroquine sulfate (Plaquenil), avoid direct sunlight, and apply protective sun cream. She did well for a while but, after a month, the morning stiffness in her fingers and knees worsened. She developed a fever of 39°C each evening accompanied by shaking chills. Enlarged lymph nodes were felt behind her ears and in the back of her neck. She also lost 4.6 kg over the course of the next 2 months.

When she returned to the hospital for a check-up, it was noted that her butterfly rash had disappeared. She had diffuse swelling of the proximal joints in her fingers and toes. Blood was drawn at this time, and the titer of anti-DNA antibodies had increased further. The serum C3 level had dropped to 46 mg dl^{-1} . Nicole was advised to take 10 mg of prednisone twice a day, as well as 250 mg of the nonsteroidal anti-inflammatory drug naproxen twice a day. This quickly controlled her symptoms, and she remained well. At her next visit, her serum C3 level was 120 mg dl^{-1} .

Immune-complex disease		
Syndrome	Autoantigen	Consequence
Subacute bacterial endocarditis	Bacterial antigen	Glomerulonephritis
Mixed essential cryoglobulinemia	Rheumatoid factor IgG complexes (with or without hepatitis C antigens)	Systemic vasculitis
Systemic lupus erythematosus	DNA, histones, ribosomes, snRNP, scRNP	Glomerulonephritis, vasculitis, arthritis

Systemic lupus erythematosus (SLE).

SLE is the typical immune-complex disease in developed countries, involving primarily women of childbearing age. Patients with SLE usually have antibodies against multiple autoantigens, including double-stranded DNA (50%), histones, and small ribonucleoproteins (30%). Autoantibodies against blood cells, such as platelets and red blood cells account for cytopenias. A third of SLE patients develop antibodies against phospholipids. When this occurs in the context of thrombotic events or recurrent fetal loss, it is diagnostic for antiphospholipid syndrome. Some anti-DNA antibodies may cross-react with brain antigen, causing the neuropsychiatric problems seen in some SLE patients. Antibodies may accumulate in patients for years (1–8 years) before patients show enough symptoms to allow for the diagnosis of SLE.

The immune complexes in SLE are small and tend to become trapped inside tissues, primarily in the kidney and, to a lesser extent, in other organs. These immune complexes fix complement and instigate injury to the kidney, joints, and other organs. Some anti-DNA antibodies are negatively charged and thus bind to basement membranes (in the kidneys), form *in situ* immune complexes, and set off inflammation.

While a few patients who have complement deficiencies in C1q (important in the clearance of immune complexes) or C4 (important in the establishment of central tolerance) develop lupus, in the majority of the patients, genome-wide association studies have revealed a genetic predisposition to lupus, indicated by the presence of several SNPs. Environmental factors (e.g., sunlight, as in this case) contribute, along with hormones (women constitute more than 90% of the patients) and aberrant immunoregulation. B cells receiving innate, cytokine, and T cell-specific help produce autoantibodies. T cells on the other hand cannot kill virally infected self cells, and do not produce the important cytokine interleukin-2, which is needed for the function of regulatory T cells and the development of cytotoxic cells. On the other hand, they produce increased quantities of the proinflammatory cytokine interleukin-17. Type I interferons (IFN- α and IFN- β), which are important in suppressing viral replication, are secreted in response to the triggering of Toll-like receptors (TLRs). Immune complexes may trigger this response in SLE. Type I IFNs promote the activation of autoreactive T cells and augment class switching and antibody production in B cells. IRF5 is a transcription factor involved in interferon synthesis, and IRF5 haplotypes were one of the first genetic susceptibility factors identified in SLE. Additional evidence for the link between IFNs and SLE is that a few patients on IFN- α treatment develop lupus. This drug-induced SLE occurs regardless of gender and tends to clear once the IFN- α is withdrawn.

The word 'lupus' is Latin for wolf, and this word is applied to a common symptom of SLE, the butterfly rash on the face. In the 19th century, the severe scarring rash on the face was named lupus because it was said to resemble the bite of a wolf. At that time, it was not possible to distinguish lupus erythematosus from lupus vulgaris, a scarring rash caused by tuberculosis. For unknown reasons, the rash is evoked by exposure to the sun (ultraviolet light). There is a seasonal variation to the onset of SLE, which is greatest in the Northern Hemisphere between March and September, when the greatest amount of ultraviolet light penetrates the atmosphere. Antimalarials such as hydroxychloroquine seem particularly helpful in the treatment of lupus skin disease.



Fig. 36.3 The so-called butterfly rash typical of systemic lupus erythematosus. Photograph courtesy of M. Walport.

Questions.

- 1 Why do you think Nicole's serum C3 was measured, both on her first visit to the hospital and after therapy?

- 2 What are the direct and indirect Coombs tests, and what did they tell us in this case?
- 3 Why was Nicole told to avoid direct exposure to sunlight?
- 4 Repeated analysis of Nicole's urine was negative. What does this mean?
- 5 Nicole had a serum IgG level of 2020 mg dL^{-1} . This substantially elevated level of IgG is commonly found in patients with SLE. How could you explain this? And what would you expect to find if we took a biopsy of Nicole's swollen lymph nodes?
- 6 The antigen in the immune complexes formed in SLE is often a complex antigen, such as part of a nucleosome or a ribonucleoprotein particle, which contains several different molecules. Patients often produce autoantibodies against each of these different components. What is the reason for the production of this variety of autoantibodies, and what type of failure in tolerance could be responsible for autoantibody production?